

Habitat suitability analysis and identification of potential fishing grounds for swordfish, *Xiphias gladius*, in the South Atlantic Ocean

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Swordfish, *Xiphias gladius*, is a highly migratory species of important commercial value and widely distributed in three oceans. Recently, the South Atlantic swordfish captured as by-catch in longline fisheries targeting tunas has contributed greatly to the overall Atlantic swordfish's landing. In this study, we have developed a habitat suitability index (HSI) model to examine the relationships between their spatio-temporal distribution and environmental factors and to identify potential fishing grounds for the swordfish in the South Atlantic Ocean using the Taiwanese distant-water longline fishery data and remote-sensing oceanographic data for 1998–2007. All the environmental factors considered – sea surface temperature (SST), mixed layer depth (MLD), sea surface height anomaly (SSHA), chlorophyll-*a* concentration (CHA) and ocean bathymetry (BAH) – were highly significant with most of the catch-per-unit-effort (CPUE) variation explained by SST. The most optimum habitat (i.e. hotspot) was found in the areas with SSTs of 27–28°C, SSHAs of –0.05 to 0.05 m, CHAs of 0.1–0.2 mg m^{–3} and BAHs of –4000 to –4500 m. The arithmetic mean model with five environmental variables was found to be the most appropriate according to the information theory based on the evaluation of different empirical HSI models in combination with different environmental factors. The bimonthly geographic information system maps of the predicted HSI values were cross-validated by the observed CPUE, suggesting that the model can be used as a tool for reliable prediction of potential fishing grounds. Because the distribution and relative abundance of swordfish are sufficiently heterogeneous in space and time, the output of this study could provide a scientific basis for time–area closures based management of this species.

1. Introduction

The swordfish, *Xiphias gladius* (Linnaeus, 1758), is a commercially important and highly migratory fish, distributed throughout the tropical, sub-tropical and temperate waters between the latitudes 45° N and 45° S (Palko *et al.* 1981). Swordfish is one of the larger predators of the oceans, reaching 455 cm in fork length (IGFA 2001) and 650 kg in weight (Nakamura 1985). Currently, the International Commission for

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the Conservation of Atlantic Tunas (ICCAT) considers the existence of three distinct management units: the North Atlantic, South Atlantic and Mediterranean Sea stocks. The results of biological (Tserpes and Tsimenides 1995), genetic (Chow and Takeyama 2000, Kasapidis *et al.* 2007) and tagging (García-Cortés *et al.* 2003, Neilson *et al.* 2007) studies clearly supported these hypotheses, although some studies have indicated exchanges between the different stocks (Alvarado *et al.* 2007).

South Atlantic swordfish are mainly caught by the longliners targeting them (e.g. Brazil and Uruguay) and tunas (e.g. Japan and Taiwan), although small catches of swordfish are also taken by other surface gear and driftnet fisheries (e.g. Angola and Ghana; ICCAT 2007). The total South Atlantic swordfish catches were relatively low before the 1980s, and then catches progressively increased to a peak of 21 780 metric tons in 1995 due to the shift in fishing effort from the North Atlantic and other oceans to the South Atlantic Ocean. During the recent decade, the South Atlantic swordfish catches followed a declining trend (e.g. 11 108 metric tons in 2008). This reduction was partly attributed, among others, to the implementation of ICCAT regulatory measures, to the shift of fleets towards other oceans and to the changes in target species (ICCAT 2007). Currently, the stock status of the South Atlantic swordfish is uncertain, resulting from the unquantified uncertainties of input parameters and the conflicting trends of catch-per-unit-effort (CPUE) indices used in the stock assessment (ICCAT 2003, 2007, 2010). The explicit effects of fish mobility and the importance of environmental factors to the distribution of fish are generally ignored in stock assessment models and fishery management policies (Kawasaki *et al.* 1991). For highly migratory species and occasions on which fishing efforts are not distributed uniformly over the species' distribution range, ignoring these movements can cause errors in the estimation of the exploitation rates (Silbert *et al.* 1999, Lehodey 2001). The habitat assessment models have become a common tool to obtain the habitat-preference information such as the functional relationships among spatial, temporal and environmental variables and fish abundance and may provide a scientific basis for habitat protection and fishery management.

Many studies have shown that environmental factors such as sea surface temperature (SST), chlorophyll-*a* concentration (CHA), mixed layer depth (MLD), sea surface height anomaly (SSHA) and ocean bathymetry (BAH) can strongly influence the distribution of swordfish and their availability to fisheries. Podestá *et al.* (1993) reported that there is an association between swordfish distribution and thermal fronts. Bigelow *et al.* (1999) indicated that swordfish CPUE of the US North Pacific longline fishery increased with latitude to a peak at 35–40° N and increased in the vicinity of temperature fronts. Temperature recordings from satellite tags have indicated that the average temperature preferred by swordfish during the day is as low as 10°C, whereas it is 28°C at night when they move up to the near-surface waters (Sedberry and Loefer 2001). The swordfish CPUE of the eastern Mediterranean Sea longline fisheries peaked at SSTs around 17°C and >26°C (Damalas *et al.* 2007). A high catch rate of swordfish was found around 22.5°C for Greek longline fleets in the eastern Mediterranean, and the swordfish tend to migrate to the preferred temperatures or areas for spawning during the peak of a spawning season (Palko *et al.* 1981, Tserpes *et al.* 2008). Furthermore, seasonal latitudinal migrations of swordfish are well documented (Nakamura 1985, Seki *et al.* 2002, Takahashi *et al.* 2003, Neilson *et al.* 2009); these migrations happen possibly as a result of seasonal changes in SST.

Swordfish are diurnal feeders rising close to the mixed surface layer at night and descending to deeper waters during the day to feed on pelagic fish and squid (Carey

1990). Although swordfish have evolved a specialized muscle that functions as a brain heater enabling them to tolerate a wide range of temperature (e.g. rapid movement between surface waters and great depths where water temperatures are much lower; Carey 1990), the vertical distribution of swordfish is generally limited by the depth of the thermocline (Block *et al.* 1992). Swordfish mainly feed on prey concentrations associated with vertical density discontinuities (Carey and Robison 1981), such as the thermocline. Draganik and Gholyst (1988) also found the highest swordfish CPUE at the thermocline. SSHA is an indicator of possible oceanographic activities such as gyres and eddies. Seki *et al.* (2002) indicated that the region fished by the Hawaii-based swordfish fleet is oceanographically referred to as the North Pacific Subtropical Frontal Zone (STFZ), which is characterized by fluctuations in SSHA. Tserpes *et al.* (2008) suggested that the SSHA range that corresponds to high swordfish density values in the eastern Mediterranean during the feeding period is found in areas surrounding eddies that are often characterized by high productivity; high concentrations of swordfish prey occur in such areas. Takahashi *et al.* (2003) also indicated that the vertical swimming behaviour of swordfish change in response to near-surface water temperatures. When the fish moved from the Kuroshio warm water area to the Kuroshio–Oyashio transition area (where the mixed layer has developed), the fish made daytime excursions from the cold deep water to the relatively warm surface. Therefore, the movement and distribution of swordfish may be influenced by oceanographic features such as sea level height and the depth of the mixed layer or thermocline.

The frontal zones and eddy fields produce locally elevated chlorophyll concentrations and zooplankton abundance, mechanically affect local aggregation of prey organisms (Owen 1981) and thereby stimulate feeding conditions (Logerwell and Smith 2001). Seki *et al.* (2002) noted that a shoaling of the vertical thermal structure of the North Pacific STFZ leads to layers of high zooplankton abundance, which is believed to concentrate swordfish, thus making it easier for the Hawaii-based long-line fishing fleet to catch them. This may suggest that it is the areas of high CHA where the preys of swordfish were most abundant due to a response to food availability (Tserpes *et al.* 2008, Yáñez *et al.* 2009). The spatial distribution and abundance of swordfish may also be determined by other factors (e.g. distinct bathymetric features) than oceanographic parameters. Many studies have indicated that the distribution of swordfish is apparently associated with bottom topographic structures (e.g. submarine canyon or hummocky bump) (Carey and Robinson 1981, Carey 1990, Podestá *et al.* 1993, Sedberry and Loefer 2001). Bathymetry at each fishing location was used as explanatory variables for evaluating the swordfish CPUE variations in the North Pacific Ocean (Bigelow *et al.* 1999) and the eastern Mediterranean Sea (Damalas *et al.* 2007).

Habitat suitability index (HSI) modelling is a valuable tool in ecology (USFWS 1980) and has also been widely used to characterize fish habitat preference, availability and quality (Rubec *et al.* 1999, Vadas and Orth 2001, Vinagre *et al.* 2006). The HSI models were based on the hypothesis that the quality of a location to a species could be determined if the biological response of the species to the range of environmental variables measured at that location was known (USFWS 1980, Eastwood *et al.* 2001). The individual biological response as a function of various environmental variables was estimated and converted into a suitability index (SI) by scaling the predicted response from 0 to 1, representing ‘poor’ to ‘good’ habitat quality, respectively (USFWS 1980, Brown *et al.* 2000). Any number of SIs could then be

combined to yield an overall description of habitat suitability. The output of HSI modelling in combination with geographic information system (GIS) technology can provide spatial and temporal information of fish habitat conditions for management decisions (Eastwood *et al.* 2001, Valavanis *et al.* 2004, Morris and Ball 2006). Recent attempts were specifically focused on how to identify and forecast potential fishing grounds for pelagic fisheries (e.g. chub mackerel, *Scomber japonicus* (Chen *et al.* 2009), and neon flying squid, *Ommastrephes bartramii* (Tian *et al.* 2009, Chen *et al.* 2010).

The objectives of this study are to develop an HSI model incorporating a GIS to detect the potential fishing grounds and to quantify the optimal ranges of key habitat variables for swordfish using the remote-sensing data in combination with fisheries data collected from the Taiwanese distant-water longline fishery in the South Atlantic Ocean. We compared the performance of different HSI models under different combinations of key habitat variables in quantifying the relationship between fish abundance indices and the predicted HSI values. The output of the spatial habitat assessment was used to draw the management implications of this species, which can help in the development of effective management schemes for the sustainable exploitation of this important fisheries resource.

2. Materials and methods

2.1 Fishery data

In this study, we focused on the South Atlantic stocks for swordfish considered by ICCAT. The Taiwanese distant-water longline fishery, extending throughout the South Atlantic Ocean (figure 1(a)), contributed 10–15% of the annual production of swordfish in the South Atlantic Ocean (ICCAT 2007). The longline catch and effort data grouped by year (1998–2007), month and $5^\circ \times 5^\circ$ grid cell were obtained from the Overseas Fisheries Development Council of Taiwan (figure 1). In addition to year and month, each catch–effort record contains information on location (latitude and longitude), fishing effort (in hooks) and catch (in number). Although the South Atlantic swordfish is a by-catch species of the Taiwanese longline fisheries, we assumed that CPUE of the longline fishery is still a good indicator of stock abundance on the fishing grounds (Maunder and Punt 2004). The nominal CPUE at i fishing grid ($5^\circ \times 5^\circ$) in month m and year y was calculated as:

$$\text{CPUE}_{y,m,i} = \frac{\sum C_{y,m,i}}{\sum E_{y,m,i}}, \quad (1)$$

where $\sum C_{y,m,i}$ is the total number of fish caught for all of the fishing vessels within i fishing grid in month m and year y , and $\sum E_{y,m,i}$ is the total number of hooks days of all fishing vessels at i fishing grid in month m and year y .

2.2 Environmental data

The following environmental variables, which were suggested in previous studies to influence swordfish abundance/distribution (see §1), were considered in this study: SST, MLD, SSHA, CHA and BAH. The environmental data were obtained from the following sources: (1) monthly SST fields on a $1^\circ \times 1^\circ$ grid for 1998–2007 from the National Centers for Environmental Prediction Reynolds optimally interpolated

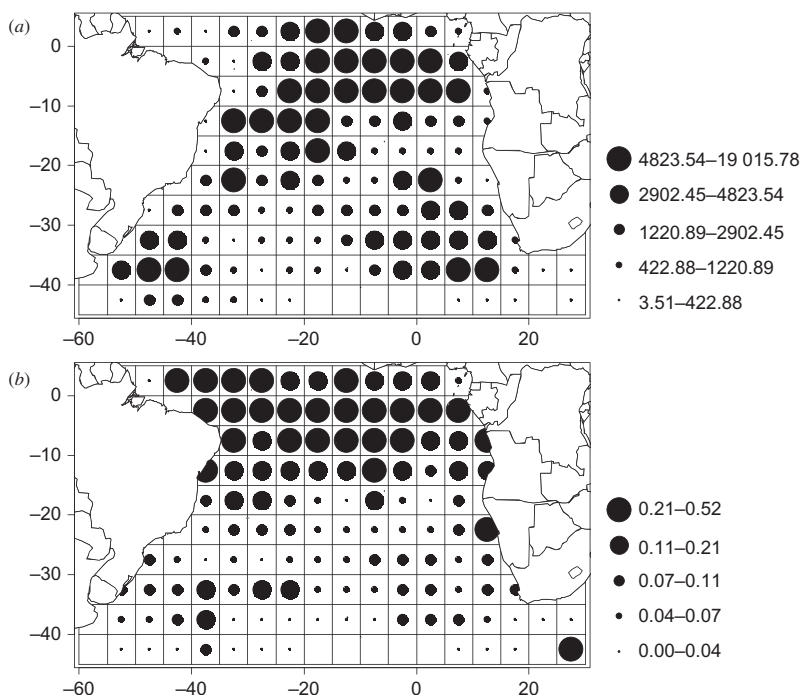


Figure 1. Spatial distribution of (a) fishing effort (1000 hooks) and (b) nominal CPUE (number of fish caught per 1000 hooks) of swordfish for the Taiwanese distant-water longline fishery in the South Atlantic Ocean from 1998 to 2007.

SST data sets (<http://poet.jpl.nasa.gov/>); (2) monthly average CHA (9 km resolution) for 1998–2007 from the Sea-viewing Wide Field-of-view Sensor (SeaWiFS, version 5.1) Level 3 mapped product (see Campbell *et al.* (1995) for the binning algorithm) (<http://oceancolor.gsfc.nasa.gov/>); (3) monthly MLD (10 min resolution) for 1998–2007 (<http://web.science.oregonstate.edu/ocean.productivity/inputMldData.php>); (4) monthly SSHA ($1/8^\circ$ resolution) for 1998–2007 (http://www7320.nrlssc.navy.mil/modas/Links/ssh_ncdef.html); and (5) the BAH information (2 min resolution) from the NOAA/NGDC/ETOPO2v2 data set (<http://www.ngdc.noaa.gov/mgg/global/etopo2.html>). For comparison with the fishing data, all oceanographic variables were converted into means by $5^\circ \times 5^\circ$ grids. The CHA values $>0.5 \text{ mg m}^{-3}$ were considered outliers and excluded from the analysis for revealing a clear relationship between CPUE and CHA.

2.3 HSI model

We first set up the SI model to identify the relationship between the data of swordfish CPUE and each environmental variable from 1998 to 2006 by the Spline smooth regression method:

$$g(\mu_i) = \mu + f(X_i), \quad (2)$$

where g is the differentiable and monotonic link function, $\mu = E(Y)$ and f is a smooth function. Cubic Spline smoothing (the software R was used to implement this analysis;

R Development Core Team 2008) was applied in the SI models. The response variable was CPUE and the environmental factor was taken as the explanatory variable. All of the SI models are based on the assumption that CPUE follows a lognormal error distribution. Examination of standardized residual plots and quantile–quantile plots provides no evidence for violation of this assumption (the results are not shown here; Maunder and Punt 2004). The predicted values of the SI model were converted or standardized into an SI, which is continuous and has values ranging from 0 to 1.0 (Tian *et al.* 2009):

$$SI = \frac{\hat{Y} - \hat{Y}_{\min}}{\hat{Y}_{\max} - \hat{Y}_{\min}}, \quad (3)$$

where $\hat{Y}$ is the predicted value of CPUE, and $\hat{Y}_{\min}$ and $\hat{Y}_{\max}$ are the minimum and maximum values of the predicted CPUE values from 1998 to 2006, respectively. The highest ($SI = 1$) and lowest ($SI = 0$) values are associated with the CPUE in a given interval of the environmental variables, which represent the most favourable and poor environmental conditions, respectively (USFWS 1980, Brown *et al.* 2000). In this study, we assumed the range of each environmental variable corresponding to SI values larger than 0.7 to be the optimal range for the South Atlantic swordfish (Tian *et al.* 2009).

The SI values derived from each environmental variable were then combined into the empirical HSI model. Three empirical HSI models (minimum model (MM), arithmetic mean model (AMM) and geometric mean model (GMM)), which are commonly used to estimate habitat availability, were considered in this study (Chen *et al.* 2009). The three empirical HSI models were described as follows:

1. MM (Van der Lee *et al.* 2006, Chen *et al.* 2009)

$$HSI_{MM} = \min(SI_1, \dots, SI_n), \quad (4)$$

2. AMM (Hess and Bay 2000, Chen *et al.* 2010)

$$HSI_{AMM} = \frac{1}{n} \sum_{i=1}^n SI_i, \quad (5)$$

3. GMM (Lauver *et al.* 2002, Chen *et al.* 2010)

$$HSI_{GMM} = \sqrt[n]{\prod_{i=1}^n SI_i}, \quad (6)$$

where SI_i is the suitability value associated with the i th environmental variable and n is the number of environmental variables considered in this study. On the basis of our previous studies on the relationship between environmental variables and swordfish CPUE (Bigelow *et al.* 1999, Sedberry and Loefer 2001, Tserpes *et al.* 2008), we considered SST as the primary variable for identifying habitat for swordfish. Furthermore, a forward selection of different SI values was used to estimate the habitat availability for each HSI model.

2.4 Model selection, habitat availability and model validation

We assumed that a positive simple linear relationship exists between the value of HSI (habitat availability) and CPUE (Chen *et al.* 2010). For evaluating the model performance in detecting the potential fishing ground of swordfish, the goodness of fit of the above linear relationships for different HSI models under various combinations of SI values was determined and compared based on the adjusted R^2 and Akaike information criterion (AIC). The model that yielded the minimum AIC value was selected as the best model (Chen *et al.* 2009, 2010).

Habitat availability was predicted by month for every $5^\circ \times 5^\circ$ grid cell in the South Atlantic Ocean for which environmental data were available to identify the spatial variation of swordfish and how they change over time. The set of cells that may be 'hotspots' for swordfish abundance was defined as those for which habitat suitability is in the top 5% (Goodyear 2003, Su *et al.* 2008). The latitude and longitude for the set of cells that were in the 'hotspot' areas derived from the best HSI model were averaged for each month during 1998–2006 to examine spatial variation and seasonal movement of 'hotspots'. Frequency distributions of habitat suitability by covariate level for the cells in the 'hotspot' areas were then used to explore which environmental conditions correspond to the preferred habitats for swordfish.

For model validation, fisheries and the corresponding environmental data from 1998 to 2006 were applied to the selected (best) HSI model. Environmental data for 2007 were used as inputs in the HSI models to estimate spatial distributions of habitat availability. The spatial distribution of HSI values predicted from the HSI model was mapped for different months (i.e. bimonth). The predicted HSI values were then compared with the observed fisheries data from the Taiwanese longline fleets in 2007 to evaluate the model performance for forecasting potential fishing grounds.

3. Results

3.1 SI of each environmental variable

The total number of observations used in the analyses is 5115, and the highest catch rates of swordfish were obtained from the tropics (figure 1(b)). The statistical analysis of SI model for each environmental variable is presented in table 1. All of the effects are statistically significant at $\alpha = 0.01$, and the correlation coefficients of the Spline smooth regression models range from 0.67 to 0.71. The SI model for SST accounted for the most deviance and had the smallest AIC value in comparison with other models (table 1). Other SI models that explained a substantial amount of the deviance ($>1\%$) were MLD, BAH and SSHA. With respect to SST and MLD, catches were

Table 1. Estimation of statistical parameters for the suitability index (SI) model of every environmental variable considered in this study.

Predictor	R	F -value	p	Deviance explained (%)	AIC
SST	0.67	66.67	<0.001	6.88	17 780
CHA	0.71	10.84	<0.001	0.38	18 224
MLD	0.70	21.56	<0.001	2.36	18 137
SSHA	0.70	18.23	<0.001	1.80	18 168
BAH	0.71	14.05	<0.001	1.21	18 197

Note: R represents Pearson correlation coefficient.

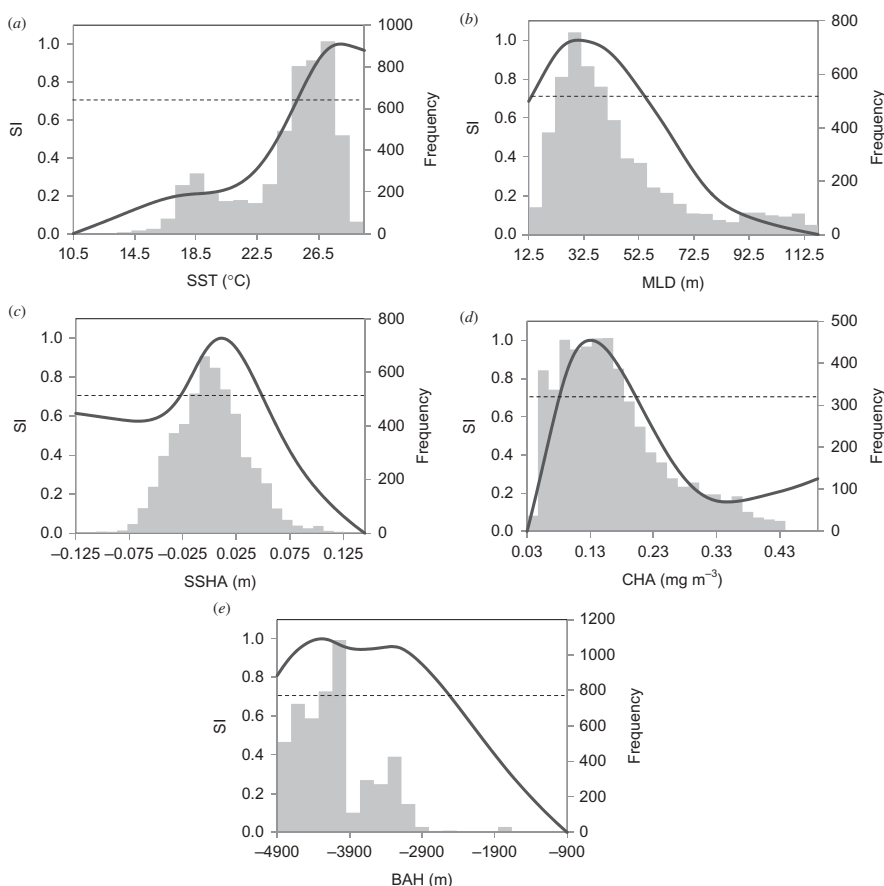


Figure 2. Suitability indices (SIs) of (a) sea surface temperature (SST), (b) mixed layer depth (MLD), (c) sea surface height anomaly (SSHA), (d) chlorophyll *a* concentration (CHA) and (e) ocean bathymetry (BAH) for the South Atlantic swordfish, *Xiphias gladius*, based on the Spline smooth regression method. The shaded histograms indicate the relative number of data points for each environmental variable. The intersections of the SI curve and the dashed line (SI = 0.7) denote the optimal range of each environmental variable.

observed in areas where SST and MLD varied from 10.50°C to 29.50°C and from 12.50 to 117.50 m, respectively. The optimal ranges of SST and MLD (i.e. SI larger than 0.7) tended to be between 24.86°C and 29.78°C (figure 2(a)) and 14.12 and 54.71 m (figure 2(b)), respectively. For SSHA, the presence of catch occurred in waters with SSHA between −0.125 and 0.145 m. The optimal range of SSHA was between −0.023 and 0.047 m (figure 2(c)). We also found catches related to CHA and BAH in waters with CHA ranging from 0.03 to 0.49 mg m^{−3} and with BAH values between −4900 and −900 m. The optimum CHA and BAH tended to be between 0.09 and 0.22 mg m^{−3} (figure 2(d)) and between −4900 and −2610 m (figure 2(e)), respectively.

3.2 Model selection

The HSI values were estimated by the three empirical models (equations (4)–(6)) for different combinations of SI values (see table 2). The different HSI models,

Table 2. Parameters of linear regression model between the CPUE and habitat suitability index (HSI) and Akaike's information criterion (AIC) for the three HSI models – minimum model (MM), arithmetic mean model (AMM) and geometric mean model (GMM) – under different combinations of environmental variables: sea surface temperature (SST), chlorophyll *a* concentration (CHA), mixed layer depth (MLD), sea surface height anomaly (SSHA) and bathymetry (BAH).

HSI model	Environmental variables	<i>a</i>	<i>b</i>	<i>p</i> (<i>F</i>)	Adjusted <i>R</i> ²	AIC	ΔAIC _w	ΔAIC _{all}
MM	+SI (SST)	−2.0664	0.6820	<0.001	0.0687	17 788	0	66
	+SI (MLD)	−2.0099	0.5657	<0.001	0.0523	17 916	127	194
	+SI (SSHA)	−2.0107	0.5674	<0.001	0.0527	17 913	125	191
	+SI (BAH)	−2.0282	0.6071	<0.001	0.0583	17 869	81	147
	+SI (CHA)	−2.0280	0.6069	<0.001	0.0583	17 869	81	147
AMM	+SI (SST)	−2.0664	0.6820	<0.001	0.0687	17 788	66	66
	+SI (MLD)	−2.0809	0.6843	<0.001	0.0623	17 838	116	116
	+SI (SSHA)	−2.2781	0.9306	<0.001	0.0683	17 792	70	70
	+SI (BAH)	−2.3591	1.0678	<0.001	0.0750	17 739	17	17
	+SI (CHA)	−2.5209	1.3227	<0.001	0.0771	17 722	0	0
GMM	+SI (SST)	−2.0664	0.6820	<0.001	0.0687	17 788	64	66
	+SI (MLD)	−1.9606	0.6123	<0.001	0.0636	17 828	104	106
	+SI (SSHA)	−1.9561	0.6834	<0.001	0.0685	17 790	66	68
	+SI (BAH)	−1.9350	0.7625	<0.001	0.0746	17 741	17	19
	+SI (CHA)	−1.9390	1.0377	<0.001	0.0768	17 724	0	2

Notes: *a* and *b* are parameters of simple linear regression. ΔAIC_w is the difference in AIC values between that model and the best model within each group. ΔAIC_{all} is the difference in AIC values between that model and the best model.

which included one to five environmental variables, were evaluated for the most parsimonious HSI model. The HSI model with one variable (SST) was the best for predicting CPUE in an area where the MM was applied (AIC = 17 788; table 2). However, the HSI models of AMM and GMM with only SST also provided moderate support of model performance (ΔAIC_w = 66 and 64). This implied that SST was the most important environmental variable in determining the habitat availability of this species. The HSI model with five variables (i.e. SST, MLD, SSHA, BAH and CHA) was the best when the AMM and GMM were applied (table 2). When all environmental variables were included for the three empirical HSI models, the AMM model yielded better results in predicting the CPUE because its AIC value was the smallest (AIC = 17 722). However, the GMM also performed well in predicting the CPUE (AIC = 17 724). In this study, we determined that the AMM with five variables, SST, MLD, SSHA, BAH and CHA, was the best HSI model (table 2).

To further compare the performances of the best two HSI models (AMM and GMM with five environmental variables), we estimated the average actual percentage of fishing effort, average actual percentage of catch and average CPUE of the South Atlantic swordfish according to the grouped HSI values by each model from 1998 to 2006 (figure 3). The preferred habitat of swordfish (HSI value >0.6) had 90.1% of the total fishing effort and 96.2% of the total catch according to the AMM, but 36.5% of total fishing effort and 57.5% of the total catch for the GMM (figures 3(a) and (b)). For the less preferred habitat (HSI value 0.2–0.6), 44.4% of the total fishing effort and 58.7% of the total catch were derived from the GMM, but moderate percentages

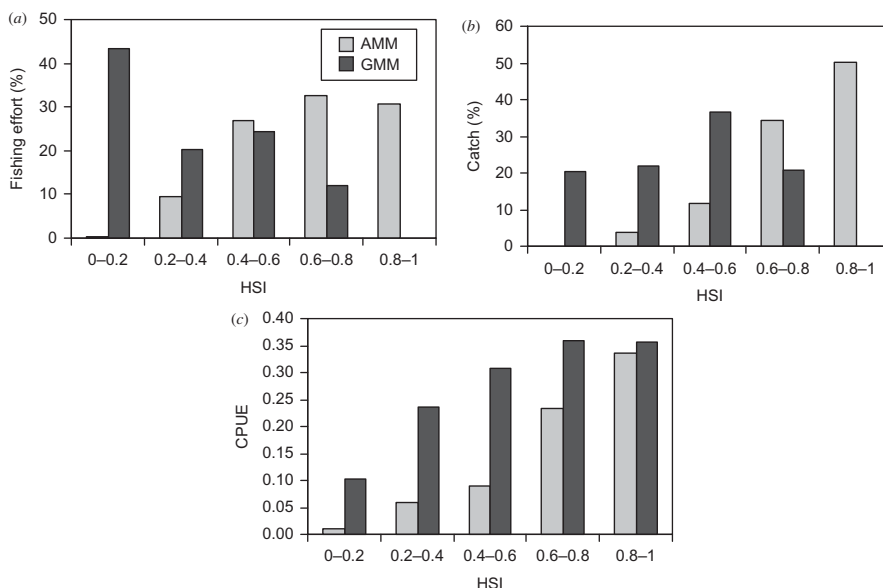


Figure 3. Relationship between the habitat suitability index (HSI) values estimated from the arithmetic mean model (AMM) and the geometric mean model (GMM) with five environmental variables and (a) percentage of fishing effort, (b) percentage of catch and (c) average catch-per-unit-effort (CPUE) (number of fish caught per 1000 hooks) in each stratum (0.2 interval) for the South Atlantic swordfish, *Xiphias gladius*, from 1998 to 2006.

of the total fishing effort (36.3%) and the total catch (15.6%) were derived from the AMM. The area with an HSI value of less than 0.2 (i.e. poor habitats) yielded 0.4% and 43.4% of the total fishing effort according to the AMM and GMM, respectively, and produced 0.01% and 20.49% in terms of the total catch.

Generally, both AMM and GMM indicated that the CPUE values increased with the strata of HSI values, but the CPUE values derived from the AMM-based HSI were well separated in comparison with the CPUE values derived from the GMM-based HSI values (figure 3(c)). When the HSI values for an area ranged from 0.6 to 0.8 and larger than 0.8, the average CPUE values were 0.23 and 0.33, respectively, according to the AMM, but the average CPUE values were 0.35 and 0.35 according to the GMM. Therefore, all of the results from the fishery data from 1998 to 2006 indicated that the AMM was more suitable than the GMM for estimating the HSI for the South Atlantic swordfish.

3.3 Hotspot analysis

There was a clear indication of seasonal latitudinal and longitudinal movement of 'hotspots' of swordfish based on the results of the AMM (figure 4). The latitudes and longitudes at which 'hotspots' occur seemed to be consistent over time, although there was variability in some months. Furthermore, the longitudinal variation of the movement of 'hotspots' (figure 4(a)) was higher than the latitudinal variation (figure 4(b)). Environmental characteristics corresponding to the cells in the 'hotspot' areas (the most preferred habitats) for South Atlantic swordfish are shown in figure 5. The 'hotspots' are predicted to occur when SST is between 27°C and 28°C, MLD is

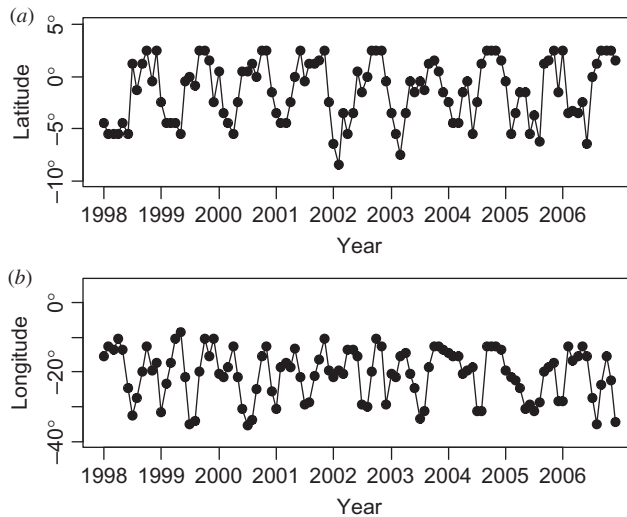


Figure 4. Average (a) latitude and (b) longitude for the spatial cells in the ‘hotspot’ areas for each month from 1998 to 2006 estimated from the arithmetic mean model with the five environmental variables for the South Atlantic swordfish, *Xiphias gladius*.

between 25 and 35 m, SSHA is between -0.05 and 0.05 m, CHA is between 0.1 and 0.2 mg m^{-3} and BAH is between -4000 and -4500 m.

3.4 Validating habitat suitability models

Bimonthly GIS map of CPUE values in 2007 was superimposed on the bimonthly HSI values estimated from the AMM based on the observed environmental data in 2007 (figure 6). Generally, higher CPUE values occurred in areas where higher HSI values were found. This implied that the selected AMM could yield a reliable prediction of the potential fishing ground for South Atlantic swordfish. There was a seasonal variation of spatial distribution of the HSI values. Most study areas were defined as the preferred habitats in the first two bimonth intervals (i.e. January–February and March–April), but the preferred habitats were concentrated in the areas between 5° and -10° N along the latitudinal direction in the fourth and fifth bimonths (i.e. July–August and September–October; figure 6).

4. Discussion

Our results indicate that each of the selected environmental variables (SST, MLD, SSHA, CHA and BAH) played an important role in explaining swordfish abundance variations. We have demonstrated the existence of non-linear relationships between the swordfish CPUE values and the environmental variables considered in this study. Of the four environmental variables considered, SST can better capture the variability of CPUE than other three variables. The ‘hotspots’ of swordfish abundance (the most optimum habitat) were concentrated around $27\text{--}28^\circ\text{C}$ isotherm (figure 5). These findings are in agreement with the observations of other studies in the Atlantic and Pacific Oceans (Nakamura 1985, Draganik and Cholyst 1988, Bigelow *et al.* 1999, Sedberry and Loefer 2001). Furthermore, the HSI models with only SST performed moderately

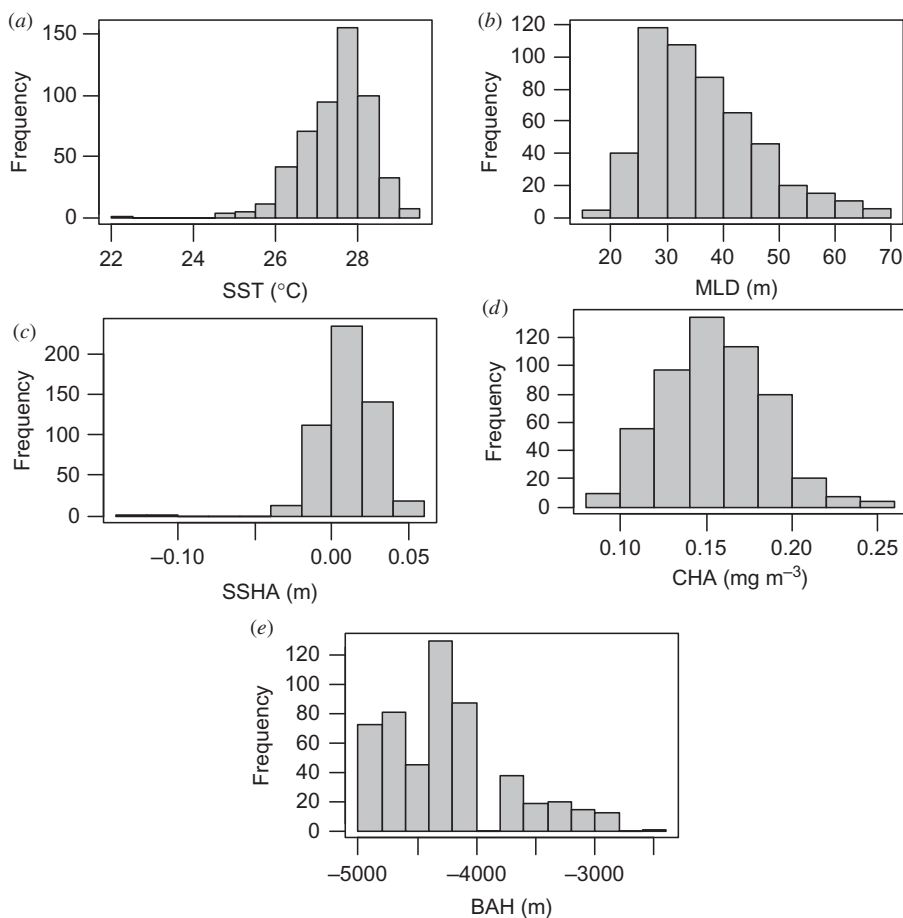


Figure 5. Frequency distribution of (a) sea surface temperature (SST), (b) mixed layer depth (MLD), (c) sea surface height anomaly (SSHA), (d) chlorophyll-*a* concentration (CHA) and (e) ocean bathymetry (BAH) for the spatial cells that are in the 'hotspot' areas based on the arithmetic mean model with five environmental variables for the South Atlantic swordfish, *Xiphias gladius*, developed using data from 1998 to 2006.

well in explaining the nominal CPUE variance (table 2). In fact, the commercially important pelagic fish have been reported to be sensitive to SST constantly and tend to have a seasonal optimum SST range (Block 1997, Bigelow *et al.* 1999, Chen *et al.* 2005, Su *et al.* 2008). Because SST data can be easily obtained in near real-time from remote-sensing data, they can be considered as basic input data in the preliminary forecasting for the potential fishing ground (Chen *et al.* 2010).

The selection of type and number of environmental variables for inclusion in the HSI modelling is critical in the successful identification of optimal habitats (Tian *et al.* 2009). On the basis of the goodness-of-fit diagnosis, we found that the HSI model with five SI variables was the best for predicting the CPUE variation when the AMM was applied (table 2). However, swordfish distribution and abundance are known to be influenced by other factors such as availability of prey, currents, thermal fronts, latitude and longitude, time of the day, season, light levels, lunar phases, dissolved

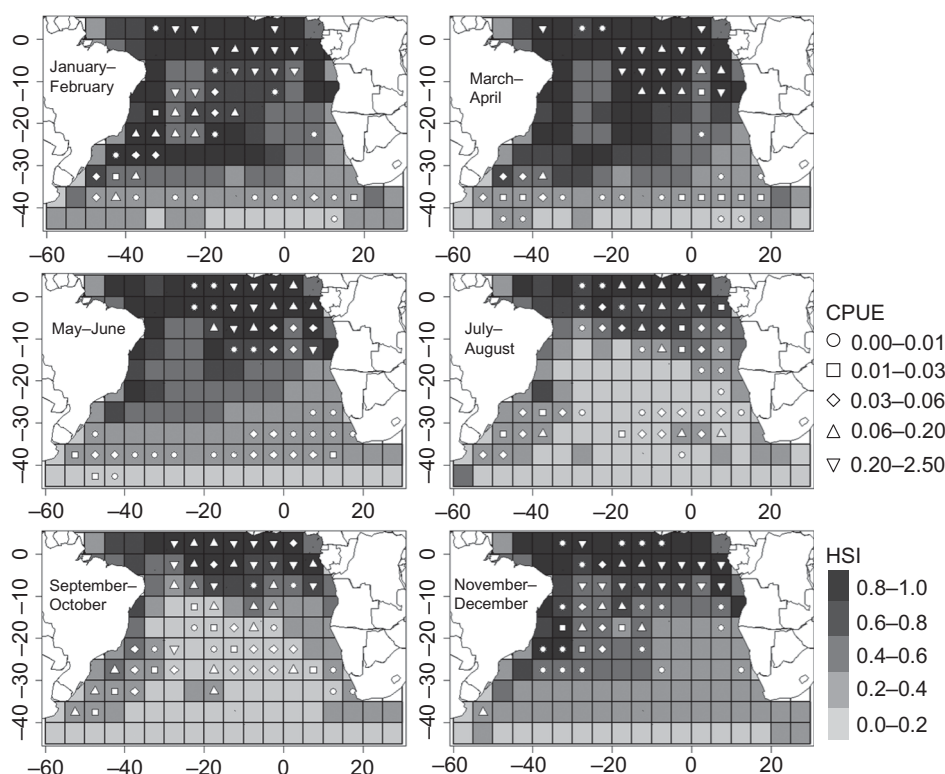


Figure 6. Bimonthly catch-per-unit-effort (CPUE) (number of fish caught per 1000 hooks) distribution for the South Atlantic swordfish, *Xiphias gladius*, caught by the Taiwanese longline fleets in 2007 overlaid on the habitat suitability index (HSI) map predicted based on the environmental data in 2007 using the arithmetic mean model with five environmental variables which were developed with data from 1998 to 2006.

oxygen concentration and wind velocity (Carey and Robison 1981, Nakamura 1985, Draganik and Cholyst 1988, Podestá *et al.* 1993, Bigelow *et al.* 1999, Sedberry and Loefer 2001, Seki *et al.* 2002, Fritsches *et al.* 2005). As a result, more factors may need to be incorporated and evaluated in future analyses; however, these variables are likely to be correlated, and therefore, the dimension reduction technique (e.g. principal component analysis) can be used to develop the environment index (Daskalov 1999).

Different methods used in processing SI models can yield different results in HSI modelling. A variety of approaches can be used to estimate the SI values based on the fishery data (fishing effort or CPUE) and environmental factors. The most common approach is the histogram method (Vinagre *et al.* 2006, Chen *et al.* 2010). However, the SI value derived from the histogram method may be affected by the number of levels for categorizing the SI values. In this study, the SI values were derived from the Spline smooth regression method and standardized to 0–1 by equation (3). With this equation, we designated the area with the higher CPUE as optimal habitat and vice versa. We chose the Spline regression method rather than the histogram method for the subsequent analysis because the Spline regression method can deal with possible non-linear relationships between covariate and response variables in a

semi-parametric manner (Maunder and Punt 2004). In addition, most of the previous studies focused on exploring the relationships between spatial and environmental factors for swordfish, and fishery catch rates were based on the extension of the Spline regression method (e.g. generalized additive models) (Bigelow *et al.* 1999, Damalas *et al.* 2007, Tserpes *et al.* 2008).

In this study, CPUE has been considered as a reliable relative abundance indicator of the swordfish in estimating SI values when commercial fishery data were used. However, many studies argue that fishermen tend to target areas where they believe fish are distributed, and thus CPUE values may be a biased indicator of spatial and temporal variations of fish population when the assumption of randomly distributed fishing efforts with respect to fish is not met (Hilborn and Walters 1992, Pedro 2006). Tian *et al.* (2009) evaluated the HSIs derived from CPUE and fishing effort data for neon flying squid in the northwestern Pacific Ocean and suggested that the HSI model according to CPUE-based SI tended to overestimate the range of optimal habitats and underestimate monthly variations in the spatial distribution of optimal habitats. Because the South Atlantic swordfish is by-caught by the Taiwanese distant-water longline fishing vessels that target tuna, CPUE might provide a better indication of swordfish abundance than fishing effort (i.e. implicitly relative abundance index for tunas). However, we may need to conduct more studies to evaluate the HSI model performance using other relative abundance indices (e.g. fishing effort or fishery-independent CPUE).

Many models were developed to quantify the relationships between fish distribution and their habitats. For example, Le Pape *et al.* (2003) quantified the distribution of common sole (*Solea solea*) in the Bay of Biscay, using a general linear model. Zainuddin *et al.* (2008) and Su *et al.* (2008) applied the generalized linear models and generalized additive models for modelling the albacore tuna (*Thunnus alalunga*) and blue marlin (*Makaira mazara*) distributions in the Pacific Ocean, respectively. Norcross *et al.* (1997) modelled habitat suitability for the juvenile flatfishes around Kodiak Island (Alaska) by using a regression tree analysis. Song and Zhou (2010) applied the quantile regression method to identify optimal habitats for the bigeye tuna (*Thunnus obesus*) in the Indian Ocean. However, the empirical HSI models have several advantages in ecological studies and are used extensively by ecologists (Brown *et al.* 2000, Vadas and Orth 2001, Ahmadi-Nedushan *et al.* 2006). For the HSI modelling, different structures of HSI models for integrating the SI values can yield different predictions of HSI values, and the integrated HSI value is also influenced by the number of environmental variables considered, the input data (relative index of abundance/biomass) and the reliability of the SI curve (Tian *et al.* 2009, Chen *et al.* 2010). When all environmental variables were included in the HSI model, we found that the AMM performed better than the other two models (MM and GMM) based on the goodness of fit of the relationship between CPUE and HSI values (table 2). A similar finding has been reported in other studies. For example, Chen *et al.* (2009) suggested the AMM as the optimum HSI model in studying habitat suitability for chub mackerel in the East China Sea. The AMM with three of the environmental variables (SST, SSHA and CHA) was also defined as the most parsimonious HSI model for neon flying squid in the northwestern Pacific Ocean.

Swordfish are a highly migratory, oceanic and pelagic species with a cosmopolitan distribution. It is hard to identify any fish species that is spatially randomly distributed. However, pelagic fish tend to move within a habitat in response to local biotic and abiotic conditions or move from one habitat to another, following an ontogenetic change

resulting from a biological requirement motivated by the need to feed and reproduce (Nakamura 1969). Bigelow *et al.* (1999) suggested that the reason why swordfish migrate seasonally is related to certain isotherms and the frontal energy in the SST in the North Pacific Ocean based on the integrated fishery and oceanographic information. Takahashi *et al.* (2003) revealed a cyclic seasonal migration of swordfish between the food-rich cold Oyashio area (40–45° N) during summer and a subtropical area (10–20° N) during winter in the northwestern Pacific Ocean based on the satellite-tag records. In the western North Atlantic Ocean, Neilson *et al.* (2009) also indicated the movement of swordfish with residence in temperate waters from June to October, followed by migration to the south into the Caribbean Sea, with fish remaining there until April based on the satellite archival tags. Tserpes *et al.* (2008) suggested that swordfish in the eastern Mediterranean Sea migrate towards the eastern Levantine basin for spawning, concentrating in specific areas during the peak of the spawning season based on the predictions of the spatial model. However, information about the movement of the South Atlantic swordfish is limited and unclear. In this study, we demonstrate that the geographic distribution of the CPUE on swordfish varies seasonally in response to their movement of certain preferred habitats. Clearly, it implies that the South Atlantic swordfish are not randomly distributed in relation to environmental conditions. The most optimum habitats are concentrated around the 27–28°C isotherm, 20–30 m MLD isobath, –0.02 to 0.03 m SSHA contour line, –4000 to –4500 m bathymetric contour and 0–0.5 mg m^{–3} CHA isopleths. Although the bulk of the optimal habitat is found in warm tropical waters (figure 6), there are also clear indications of changes in the preferred latitude and longitude seasonally (figure 5). Overall, the frontal structures and eddy fields may play an important role in the development of potential swordfish habitats by stimulating the feeding conditions. These habitats account for significant biophysical responses for swordfish to find prey abundance and physiologically suitable conditions (Podestá *et al.* 1993, Seki *et al.* 2002). We suggest that the identification of oceanic front through chlorophyll and SST (Belkin *et al.* 2009, Belkin and O'reilly 2009) or SSH (Sokolov and Rintoul 2002) satellite maps and investigation of the relationship between oceanic front and swordfish habitat could improve the detecting and predicting potential for swordfish fishing grounds.

Effective management and conservation of the migratory pelagic fish in the open oceans depend on determining the spatial distribution of fish, fisheries and the ecological system (Sibert and Hampton 2003, Su *et al.* 2008). The information derived from the HSI models can be used in identifying key habitats of the South Atlantic swordfish and help fishermen in locating optimal fishing grounds and reducing the searching time the fishermen invest. Because most of the fishing mortality on the billfish species results from by-catch in the pelagic longline fisheries that target other species, particularly tunas, spatial–temporal closures were suggested for species such as billfish that have pelagic larvae and highly migratory adults (Alverson *et al.* 1994, Goodyear 1999, Hobday and Hartmann 2006). The main objective of using time–area strategies is to take advantage of variation in the degree of co-occurrence between target and by-catch species (Murawski 1992). Therefore, the temporal and spatial habitat preference patterns and the time-consistent ‘hotspot’ areas identified in this article could provide a scientific basis for the management of this species. It should be noted that the effectiveness of such controls obviously depends on the degree of overlap between the by-catch and the targeted species (Adlerstein and Trumble 1992). We suggest further evaluation of the empirical HSI models with respect to various sources of uncertainty and development of a dynamic near real-time habitat model incorporating

more biological and oceanographic variables, such as availability of prey, currents and fronts, and other environmental variables in finer scales for both target and by-catch species, before final recommendation and implementation for the management of the South Atlantic swordfish.

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